

Second Year B. Tech (Computer Engineering)

Subject Name and code: Computer Architecture and Organization 23UCOPCL2302

Session 2026-27

Semester-III

Class: Second Year

Poster Presentation

(Max. Marks: 10)

All students are informed to prepare and present Animated Poster Presentation during the scheduled TAE-II session.

The solution must be developed using AI Tool.

Instructions:

- 1. Poster will be well designed and animated.**
- 2. The presentation should be informative, creative, and visually appealing.**
- 3. Students must mention the name of the AI tool(s) used and explain how they were used.**

Unit No.	Problem statement	Marks	CO Mapped	BL
3	56 Analyze the memory requirements of a smartphone and design a suitable memory hierarchy considering speed, capacity, and cost. A computer system experiences slow execution despite having 57 sufficient RAM. Analyze how the memory hierarchy can affect system performance and suggest improvements. A computer uses cache, RAM, SSD and external storage. Analyze 58 the role of each level and explain how data moves through the hierarchy during program execution. Design a memory chip organization for a 4K × 8-bit memory 59 system. Determine the required address and data lines and represent the organization using a block diagram. A memory system must support 8K memory locations with 16-bit 60 words. Analyze the required address/data lines and propose a suitable chip organization. Design an appropriate RAM organization for an embedded system 61 with limited cost and power consumption. Justify the selected organization. A digital system requires non-volatile memory for storing 62 firmware. Compare ROM, PROM, EPROM and EEPROM and select a suitable technology for the application.	10M	CO3	L3

63	A processor is spending excessive time accessing main memory. Analyze how cache memory can reduce the average memory access time and propose a suitable cache arrangement. A processor uses direct-mapped cache memory. Given a sequence of memory addresses, analyze cache hits and misses and demonstrate the mapping process. Design an associative cache for a system where fast data retrieval is critical. Explain the address comparison process and analyze its performance. A processor requires a balance between cache speed and storage capacity. Analyze whether set-associative mapping is suitable and demonstrate its operation with an example. A system designer must select a cache mapping technique. Analyze direct, associative and set-associative mapping for a given workload and compare their performance characteristics. A processor has a high cache miss rate. Analyze the possible causes and propose techniques to improve the cache hit ratio. A multi-user system frequently updates cached data. Analyze suitable cache write policies and replacement strategies for maintaining performance and data consistency. Design a three-level cache hierarchy for a modern processor. Analyze the role of L1, L2 and L3 caches in improving processor performance. A computer needs to execute programs larger than the available physical RAM. Analyze how virtual memory can solve this problem and explain its benefits. A virtual address must be converted into a physical address. Demonstrate the address translation process using a suitable page-table example. A system has limited physical memory but must support multiple large applications. Design a paging-based memory management solution and explain its operation. Given a virtual address, page size and page table, calculate the corresponding physical address. Represent the complete translation process diagrammatically. A multimedia application requires memory regions with different sizes and protection levels. Analyze how segmentation can be used to manage these regions. Compare paging and segmentation for a multitasking operating system. Analyze their advantages, limitations and suitability for different applications. A processor frequently accesses the same pages and suffers from slow address translation. Analyze how a TLB can improve address translation performance. Given TLB hit ratio, TLB access time and memory access time,			
----	---	--	--	--

	calculate the effective memory access time and analyze the impact of TLB performance. A program generates a page fault during execution. Analyze the 79 sequence of events from page-fault detection to loading the required page into memory. A system has limited physical frames and frequent page faults. 80 Analyze suitable page replacement algorithms to reduce page-fault frequency. Given a page-reference string and a fixed number of frames, apply FIFO page replacement and calculate the number of page faults. 81 Given a page-reference string and available memory frames, apply Optimal Page Replacement and analyze the resulting page-fault rate. 82 A system experiences frequent page faults because of changing program locality. Apply the LRU algorithm to a given reference string and analyze its performance. 83 For a given page-reference string, apply FIFO, LRU and Optimal algorithms. Compare the number of page faults and analyze the behavior of each algorithm. 84 A system using FIFO page replacement shows an unexpected increase in page faults after adding more frames. Investigate this behavior and explain it using Belady's anomaly. 85 A computer system must transfer large amounts of data between processor and memory. Analyze the suitability of parallel bus architecture for this application. 86 A system requires reliable data communication using fewer physical connections. Analyze how serial bus architecture can address this requirement. 87 A system designer must choose between parallel and serial communication. Analyze both architectures based on speed, wiring complexity, distance and cost. 88 Design the system bus of a basic computer. Show how address, data and control buses coordinate communication among CPU, memory and I/O devices. 89 A data-intensive application requires large-capacity storage. 90 Analyze suitable mass-storage technologies considering capacity, access time, reliability and cost. A disk-based system experiences long access times. Analyze disk organization and identify factors affecting disk access performance. 91 Design a storage solution using HDD for a large data archive. 92 Analyze its organization, capacity, access mechanism and limitations. A server requires high-speed storage with low access latency. 93 Analyze SSD architecture and explain why it may be suitable for the application. A system administrator must select between HDD and SSD. 94 Analyze both technologies for operating-system storage, gaming and			
--	---	--	--	--

	large-scale data storage applications. 95 A portable device requires compact, low-power and non-volatile storage. Analyze flash memory architecture and explain its suitability. A storage designer must choose between NAND and NOR flash. 96 Analyze their organization, performance and applications and justify their use in different systems. 97 Design the secondary storage organization for a computer system that needs high capacity, reliability and efficient data access. A computer must communicate with multiple peripheral devices. 98 Analyze the role of an I/O system and design an appropriate I/O organization. 99 A processor must transfer a block of data to an I/O device. Analyze programmed I/O and demonstrate the complete data-transfer sequence. An industrial monitoring system must respond quickly when an 100 external sensor generates data. Analyze how interrupt-driven I/O can improve system responsiveness. Compare programmed I/O and interrupt-driven I/O for a real- 101 time monitoring system. Analyze CPU utilization, response time and data-transfer efficiency. A high-speed device needs to transfer a large block of data 102 directly to memory. Analyze how DMA can reduce CPU involvement. Design a DMA controller for a system containing CPU, memory 103 and an I/O device. Explain the signals and steps involved in DMA transfer. A system must support different types of DMA transfers. Analyze 104 DMA transfer modes and select an appropriate mode for a given application. A high-speed data acquisition system must transfer large data 105 blocks efficiently. Analyze Programmed I/O, Interrupt-Driven I/O and DMA and determine the effect on CPU utilization. Design an I/O interface for connecting a peripheral device to a 106 processor. Identify the required registers, control signals and communication mechanism. A processor receives simultaneous requests from multiple I/O 107 devices. Analyze interrupt handling and propose a suitable mechanism for managing multiple interrupts. Design an integrated memory and I/O organization for an 108 embedded system. Analyze the interaction among CPU, cache, main memory, secondary storage and I/O devices.			
--	---	--	--	--

Rubrics for evaluation (Total evaluation marks 10)

Sr. No.	Indicator	Marks	Criteria (Rubric Levels)
1	Technical content & Concept explanation with real-world application	4	4 – Correct, optimized, well-structured 3 – Mostly correct with minor issues 2 – Partial 1 – Major errors
2	Animation/visualization quality,Originality & creativity	3	3 –Clear explanation and AI role 2 – Poor Visualizations 1 –Weak explanation 0 – No creativity
3	Presentation, communication & Q&A	3	3 –Clear concept explanation and AI role 2 –Partial explanation 1 –Weak explanation 0 – No explanation

Subject Teacher

HOD